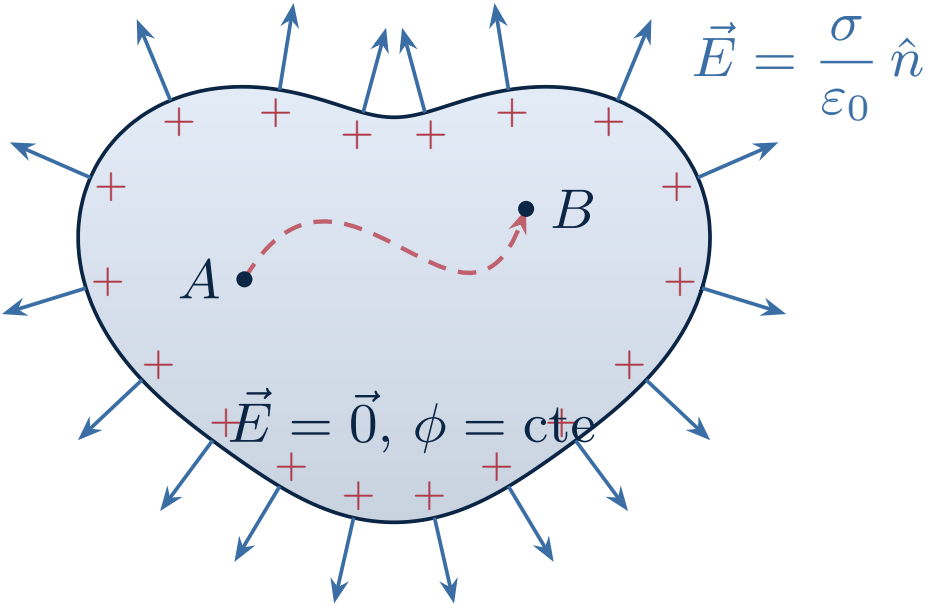
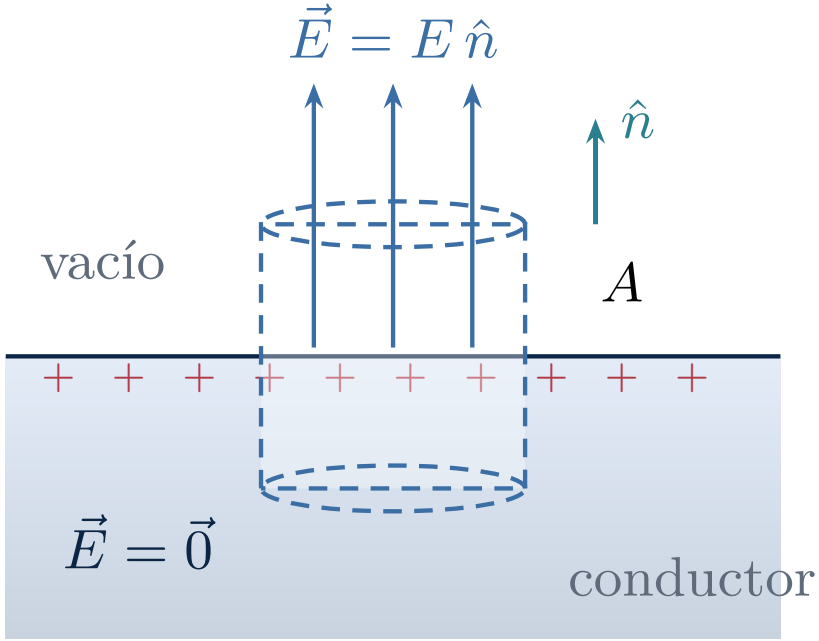


$$\phi_B - \phi_A = -\int_A^B \vec{E}\cdot \mathrm{d}\vec{\ell} = 0$$

(a)



$$\oiint \vec{E}\cdot \mathrm{d}\vec{S} = EA = \frac{\sigma A}{\varepsilon_0}$$

(b)